



2009

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

MATHEMATICS EXTENSION 2

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using blue or black pen
- Board approved calculators may be used
- Write your name on each page
- Each question is to be started on a new page.
- This examination paper must NOT be removed from the examination room
- There is a total of eight questions.
- Each question is worth 15 marks.
- Marks may be deducted for careless or badly arranged work.

Question 1 (15 marks) Start a NEW page.**Marks**

a) Find $\int \cos x \sin^4 x \, dx$. 1

b) Find $\int \frac{dx}{x^2 - 4x + 8}$. 2

c) Use the substitution $u = x - 2$ to find the exact value of $\int_1^3 x(x-2)^5 \, dx$. 3

d) i) Find the values of A , B and C so that 2

$$\frac{5}{(x^2 + 4)(x + 1)} \equiv \frac{Ax + B}{x^2 + 4} + \frac{C}{x + 1}.$$

ii) Hence find $\int \frac{5}{(x^2 + 4)(x + 1)} \, dx$. 3

e) i) If $I_n = \int_1^e x(\ln x)^n \, dx$ for $n = 0, 1, 2, 3, \dots$ use integration by parts 2

to show that $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$ for $n = 1, 2, 3, \dots$

ii) Hence find the value of I_2 . 2

Question 2 (15 marks) Start a NEW page.**Marks**a) If $z = 1 - i$, find

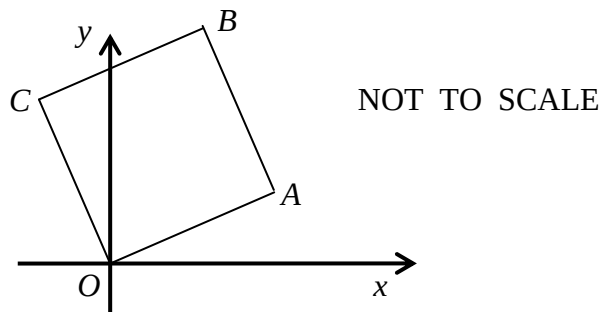
i) $\bar{z}$ 1

ii) $|z|$ 1

iii) $\arg z$ 1

iv) $\arg iz$ 1

v) z^6 in $a + ib$ form. 2

b) Express $\frac{i^5(1-i)}{2+i}$ in the form $a + ib$ where a and b are real. 3c) Graph the region in the Argand diagram which simultaneously satisfies $1 \leq |z - i| \leq 2$ and $\operatorname{Im} z \geq 0$. 3d) In the Argand diagram A represents the point $z_1 = \sqrt{3} + i$, and O is the origin.Given that $OABC$ is a square:i) Find the complex number (z_3) represented by C . 1ii) Find the complex number (z_2) represented by B . 2

Question 3 (15 marks) Start a NEW page.**Marks**

- a) i) Without using calculus, draw a good size neat sketch of $y = (x+1)^2(1-x)$. 2
- ii) On a separate diagram using the same scale as above, and also without calculus, sketch $y^2 = (x+1)^2(1-x)$, paying close attention to the shape of the curve as 'y' approaches zero. 2
- b) Neatly sketch each of the following on separate axes for $0 \leq x \leq 2\pi$.
- i) $y = \sin^2 x$. 1
- ii) $y = |\sin x|$. 1
- iii) $y = \sqrt{\sin x}$. 1
- iv) $y = \frac{1}{\sin x}$. 1
- v) $y = \frac{|\sin x|}{\sin x}$. 1
- vi) $y = e^{\sin x}$. 2
- c) A plane curve is defined by the equation $x^2 + 2xy + y^5 = 4$. 4
- The curve has a horizontal tangent at the point $P(X,Y)$.
- By using implicit differentiation or otherwise, show that X is the unique solution to the equation $X^5 + X^2 + 4 = 0$.
- [Do not solve this equation]

Question 4 (15 marks) Start a NEW page.**Marks**

- a) Given the hyperbola $16x^2 - 9y^2 = 144$, find:
- | | |
|--------------------------------------|---|
| i) the length of the transverse axis | 1 |
| ii) the eccentricity | 1 |
| iii) the coordinates of the foci | 1 |
| iv) the equations of the directrices | 1 |
| v) the equations of the asymptotes. | 1 |
- b) If α, β and γ are the roots of the equation $x^3 + 6x + 1 = 0$, find the polynomial equation whose roots are $\alpha\beta, \beta\gamma$ and $\alpha\gamma$. 3
- c) Consider the equation $x^4 - 5x^3 + 7x^2 + 3x - 10 = 0$.
- | | |
|---|---|
| i) Given that $2 - i$ is a root of the equation explain why $2 + i$ is also a root. | 1 |
| ii) Find the other roots of the equation. | 3 |
- d) The area enclosed between the curves $y = \sqrt{x}$ and $y = x^2$ is rotated about y - axis through one complete revolution. Use the method of cylindrical shells to find the volume of the solid that is generated. 3

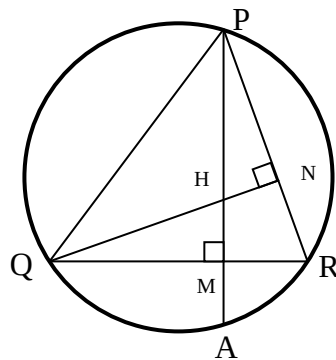
Question 5 (15 marks) Start a NEW page.**Marks**

a) Solve for x : $2^{3x+1} = 5^{x+1}$, correct to 3 significant figures. 2

b) i) Prove the result $\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$ 2

ii) Hence use the above result to evaluate $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{1 + \sin x} dx$ 3

- c) The altitudes PM and QN of an acute angled triangle PQR meet at H . PM produced cuts the circle PQR at A . [A larger diagram is included, use it and submit it with your solutions]



- i) Explain why PQMN is a cyclic quadrilateral. 1
- ii) Prove that $HM = MA$. 3

- d) The point $P(a\cos\theta, b\sin\theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The tangent at P cuts the y -axis at B and M is the foot of the perpendicular from P to the y -axis.

- i) Show that the equation of the tangent to the ellipse at the point P is 2

given by $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$

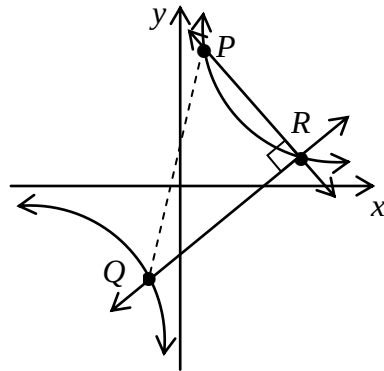
- ii) Show that $OM \cdot OB = b^2$, where O is the origin. 2

Question 6 (15 marks) Start a NEW page.

Marks

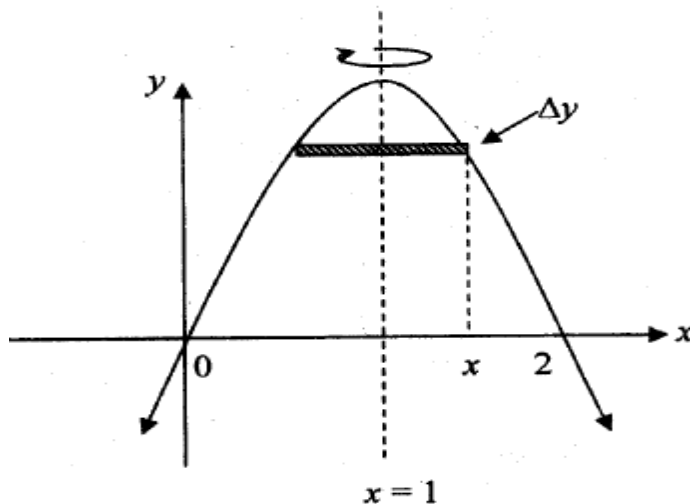
- a) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ lie on the rectangular hyperbola $xy = c^2$. 4

The chord PQ subtends a right angle at another point $R\left(cr, \frac{c}{r}\right)$ on the hyperbola.



Show that the normal at R is parallel to PQ .

- b) The area bounded by the curve $y = 2x - x^2$ and the x -axis is rotated through 180° about the line $x = 1$.



- i) Show that the volume, ΔV , of a representative horizontal slice of width Δy is given by $\Delta V = \pi(x - 1)^2 \Delta y$. 2

- ii) Hence show that the volume of the solid of revolution is given by 2

$$V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^1 \pi(1 - y) \Delta y$$

- iii) Hence, find the volume of the solid of revolution. 2

Question 6 continued on page 8

<u>Question 6 continued</u>		Marks
c)	i) Show that $a^2 + b^2 > 2ab$, where a and b are distinct positive real numbers.	1
	ii) Hence show that $a^2 + b^2 + c^2 > ab + bc + ca$, where a , b and c are distinct positive real numbers.	2
	iii) Hence, or otherwise prove that $\frac{a^2b^2 + b^2c^2 + c^2a^2}{a + b + c} > abc$, where a , b and c are distinct positive real numbers.	2

End of Question 6

Question 7 (15 marks) Start a NEW page.**Marks**

- a) Given that $P(x) = 3x^3 - 11x^2 + 8x + 4$ has a double root, fully factorise $P(x)$. 3
- b) Show that $\tan^{-1}x > x - \frac{1}{3}x^3$ for all values of $x > 0$. 3
- c) The acceleration of a particle which is moving along the x-axis is given by
- $$\frac{d^2x}{dt^2} = 2x^3 - 10x.$$
- i) If the particle starts at the origin with velocity u show that its velocity v is given by $v^2 - u^2 = x^4 - 10x^2$. 2
- ii) If $u = 3$ show that the particle oscillates within the interval $-1 \leq x \leq 1$. 4
- iii) Is the motion referred to in (ii) an example of simple harmonic motion? 1
Give a clear reason for your answer.
- iv) If $u = 6$, carefully describe the motion. 2

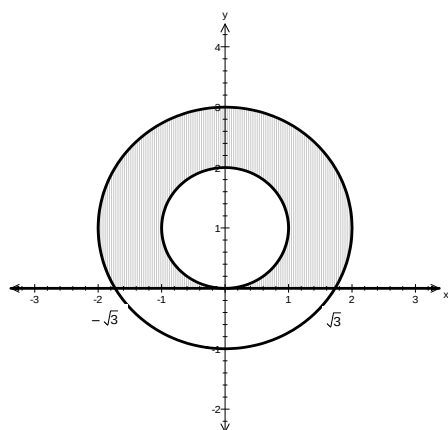
Question 8 (15 marks) Start a NEW page.**Marks**

- a) Let ω be one of the non-real cube roots of unity.
- i) Show that $1 + \omega + \omega^2 = 0$. 1
- ii) Hence find the value of $(2 - \omega)(2 - \omega^2)(2 - \omega^4)(2 - \omega^5)$. 2
- b) i) By using the expansion for $\cos(A + B)$, show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. 2
- ii) By using the substitution $x = 2\cos\theta$, solve the equation $x^3 - 3x = \sqrt{2}$. 3
- iii) Hence explain why $\cos\left(\frac{\pi}{12}\right) + \cos\left(\frac{7\pi}{12}\right) + \cos\left(\frac{9\pi}{12}\right) = 0$ 1
- c) The equation $x^3 - 3px + q = 0$, where $p > 0, q \neq 0$ are both real, has three distinct, non-zero real roots.
- i) Show that the graph of $y = x^3 - 3px + q$ has a relative maximum value of $q + 2p\sqrt{p}$ and a relative minimum of $q - 2p\sqrt{p}$. 3
- ii) Hence show giving reasons that $q^2 < 4p^3$. 3

END OF EXAM

YEAR 12 EXTENSION 2 CARINGBAH	HIGH SCHOOL TRIAL HSC SOLUTIONS 2009
1a) $\int \cos x \sin^4 x \, dx = \frac{\sin^5 x}{5} + C$ ✓ b) $\int \frac{dx}{x^2 - 4x + 8} = \int \frac{dx}{(x-2)^2 + 2^2}$ ✓ $= \frac{1}{2} \tan^{-1} \left(\frac{x-2}{2} \right) + C$ ✓ c) Let $u = x - 2 \rightarrow x = u + 2 \rightarrow du = dx$ When $x = 1, u = -1; x = 3, u = 1$ $\therefore I = \int_{-1}^1 (u+2) \cdot u^5 \, du$ ✓ $= \int_{-1}^1 u^6 + 2u^5 \, du$ $= \left[\frac{u^7}{7} + \frac{2u^6}{6} \right]_{-1}^1$ ✓ $= \left(\frac{1}{7} + \frac{1}{3} \right) - \left(-\frac{1}{7} + \frac{1}{3} \right) = \frac{2}{7}$ ✓ d) i) $5 \equiv (Ax + B)(x + 1) + C(x^2 + 4)$ Let $x = -1 : 5 = 5C \rightarrow C = 1$ ✓ Let $x = 0 : 5 = B + 4C \rightarrow B = 1$ Let $x = 1 : 5 = 2(A + B) + 5C \rightarrow A = -1$ ✓ ii) $I = \int \frac{-x+1}{x^2+4} + \frac{1}{x+1} \, dx$ $= \int \frac{-x}{x^2+4} + \frac{1}{x^2+4} + \frac{1}{x+1} \, dx$ ✓ $= -\frac{1}{2} \ln(x^2 + 4) + \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + \ln(x + 1) + c$ ✓✓ e) $I_n = \int_1^e x (\ln x)^n \, dx$ Let $u = (\ln x)^n, \quad v' = x$ $u' = \frac{n}{x} (\ln x)^{n-1}, \quad v = \frac{x^2}{2}$ ✓	$\therefore I_n = \left[\frac{x^2}{2} (\ln x)^n \right]_1^e - \int_1^e \frac{x^2}{2} \cdot \frac{n}{x} (\ln x)^{n-1} \, dx$ $= \frac{e^2}{2} - \frac{1^2}{2} \cdot 0^n - \frac{n}{2} I_{n-1} = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$ ✓ e)ii) $I_2 = \frac{e^2}{2} - \frac{2}{2} I_1$ $= \frac{e^2}{2} - \left[\frac{e^2}{2} - \frac{1}{2} I_0 \right]$ ✓ $= \frac{1}{2} \int_1^e x \, dx = \frac{e^2 - 1}{4}$ ✓ 2a) i) $\bar{z} = 1 + i$ ✓ ii) $z = \sqrt{2}$ ✓ iii) $\arg z = -\frac{\pi}{4}$ ✓ iv) $\arg iz = \arg(i) + \arg z = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$ ✓ v) $z^6 = \left(\sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4} \right) \right)^6$ ✓ $= \sqrt{2}^6 \cdot \operatorname{cis} \left(-\frac{6\pi}{4} \right) \text{ (DMT)}$ $= 8 \times i = 0 + 8i$ ✓ b) $\frac{i^5(1-i)}{2+i} = \frac{i \cdot (i^2)^2(1-i)}{2+i} \times \frac{2-i}{2-i}$ ✓ $= \frac{i(1-i)(2-i)}{1+4}$ ✓ $= \frac{i(2-i-2i-1)}{5}$ $= \frac{3}{5} + \frac{1}{5}i$ ✓

2c)

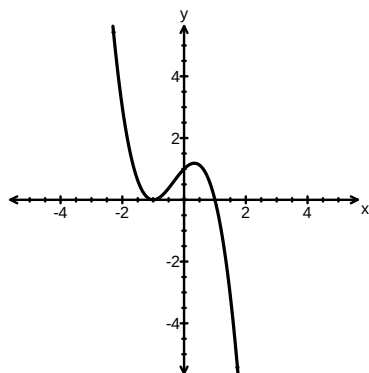


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d) i) C is $z_3 = i(\sqrt{3} + i) = -1 + i\sqrt{3}$ ☑

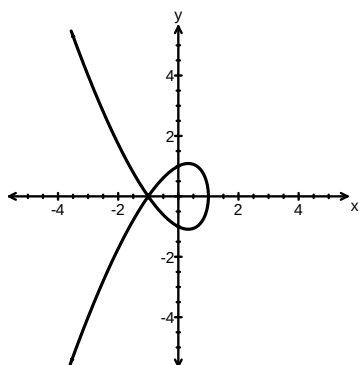
ii) B is $z_2 = \overrightarrow{OA} + \overrightarrow{AB}$
 $= \overrightarrow{OA} + \overrightarrow{OC}$ ☑
 $= z_1 + z_3$
 $= (\sqrt{3} - 1) + i(1 + \sqrt{3})$ ☑

3a) i)



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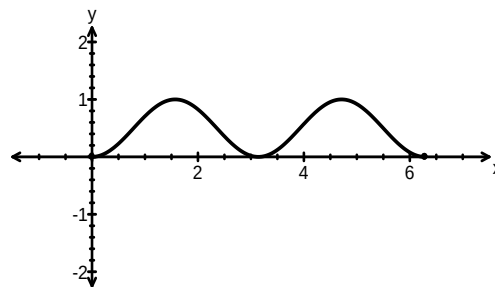
ii)



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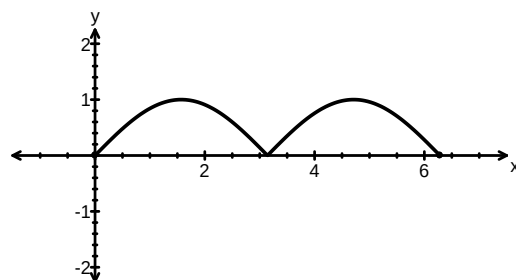
3b) i)

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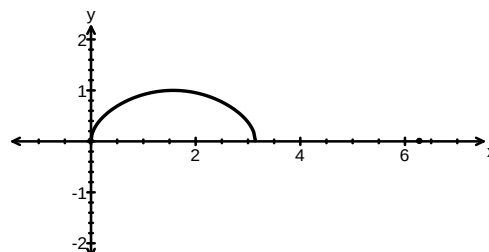
ii)

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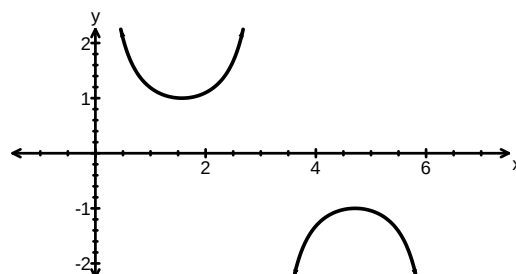
iii)

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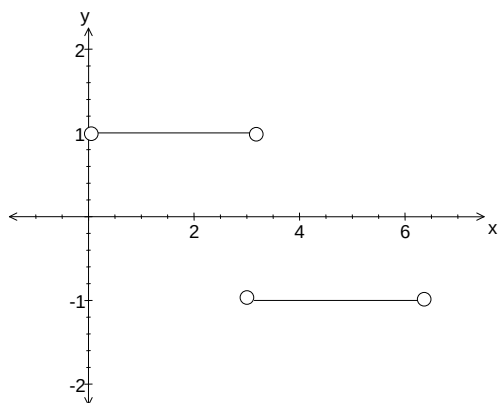


iv)

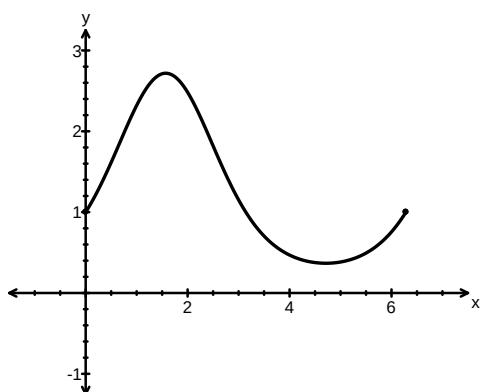
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3b) v)



vi)



c) Horizontal tangent at $x = X \rightarrow \frac{dy}{dx} = 0$

$$\frac{d}{dx}(x^2 + 2xy + y^5) = \frac{d}{dx}(4)$$

$$2x + 2\left(y \frac{dy}{dx} + y^4 \frac{dy}{dx}\right) = 0$$

$$(2x + 5y^4) \frac{dy}{dx} = -2(x + y)$$

$$\therefore \frac{dy}{dx} = \frac{-2(x + y)}{(2x + 5y^4)}$$

$$\therefore -2(x + y) = 0 \rightarrow y = -x \text{ where } x = X$$

$$\therefore X^2 + 2X(-X) + (-X)^5 = 4$$

$$X^2 - 2X^2 - X^5 = 4$$

$$\therefore X^5 + X^2 + 4 = 0$$

$$4a) \quad \frac{x^2}{9} - \frac{y^2}{16} = 1 \rightarrow a = 3, b = 4$$

$$i) \text{ transverse axis} = 6$$

$$ii) e^2 = 1 + \frac{b^2}{a^2} = \frac{25}{9}$$

$$\therefore e = \frac{5}{3}$$

$$iii) ae = 3 \times \frac{5}{3} = 5$$

$$\therefore \text{Foci are } S(5,0) \text{ and } S'(-5,0)$$

$$iv) \frac{a}{e} = \frac{3}{5/3} = \frac{9}{5}$$

$$\therefore \text{directrices are } x = \pm \frac{9}{5}$$

$$v) \frac{b}{a} = \frac{4}{3} \rightarrow \text{asymptotes } y = \pm \frac{4}{3}x$$

b) Let the roots be: $\alpha\beta, \beta\gamma, \alpha\gamma$

which is the equivalent of: $\frac{\alpha\beta\gamma}{\gamma}, \frac{\beta\gamma\alpha}{\alpha}, \frac{\alpha\gamma\beta}{\beta}$

and since $\alpha\beta\gamma = -\frac{d}{a} = -1$

then the roots are $-\frac{1}{\alpha}, -\frac{1}{\beta}, -\frac{1}{\gamma}$.

Let $x = -\frac{1}{\alpha} \rightarrow \alpha = -\frac{1}{x}$

and since α satisfies the original equation then we get

$$\left(-\frac{1}{x}\right)^3 + 6\left(-\frac{1}{x}\right) + 1 = 0$$

which gives $x^3 - 6x^2 - 1 = 0$.

c) i) Since the coefficients are real, the complex roots occur in conjugate pairs.

ii) Let $P(x) = (x - (2 - i))(x - (2 + i))Q(x)$

$$= (x^2 - 4x + 5)(x^2 + ax + b)$$

$$\therefore ax^3 - 4x^3 = -5x^3 \rightarrow a = -1$$

$$\text{also } 5b = -10 \rightarrow b = -2$$

$$\therefore P(x) = (x^2 - 4x + 5)(x^2 - x - 2)$$

$$= (x^2 - 4x + 5)(x - 2)(x + 1)$$

Hence the other roots are $x = 2$ and $x = -1$.

$$4d) V_s = 2\pi xy \delta x$$

$$\therefore V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi xy \delta x \quad \checkmark$$

$$= 2\pi \int_0^1 x(\sqrt{x} - x^2) dx \quad \checkmark$$

$$= 2\pi \int_0^1 x^{\frac{3}{2}} - x^3 dx$$

$$= 2\pi \left[\frac{2x^{\frac{5}{2}}}{5} - \frac{x^4}{4} \right]_0^1$$

$$= \frac{3\pi}{10} u^3 \quad \checkmark$$

$$5a) \ln(2^{3x+1}) = \ln(5^{x+1})$$

$$\therefore (3x+1)\ln 2 = (x+1)\ln 5 \quad \checkmark$$

$$3x\ln 2 + \ln 2 = x\ln 5 + \ln 5$$

$$x(3\ln 2 - \ln 5) = \ln 5 - \ln 2$$

$$x = \frac{\ln 5 - \ln 2}{3\ln 2 - \ln 5} \approx 1.95 \quad \checkmark$$

$$b) i) I = \int_{-a}^a f(x) dx \\ = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

$$\text{Let } x = -u \rightarrow dx = -du$$

$$\text{When } x = -a, u = a; x = 0, u = 0 \quad \checkmark$$

$$\therefore I = -\int_a^0 f(-u) du + \int_0^a f(x) dx$$

$$= \int_0^a f(-u) du + \int_0^a f(x) dx$$

$$= \int_0^a f(-x) dx + \int_0^a f(x) dx \quad \checkmark$$

$$= \int_0^a [f(x) + f(-x)] dx$$

ii)

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{1+\sin x} dx = \int_0^{\frac{\pi}{4}} \frac{1}{1+\sin x} + \frac{1}{1+\sin(-x)} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{1+\sin x} + \frac{1}{1-\sin x} dx \quad \checkmark$$

$$= \int_0^{\frac{\pi}{4}} \frac{1 - \sin x + 1 + \sin x}{1 - \sin^2 x} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{2}{\cos^2 x} dx \quad \checkmark$$

$$= 2 \int_0^{\frac{\pi}{4}} \sec^2 x dx$$

$$= 2 \left[\tan x \right]_0^{\frac{\pi}{4}} = 2 \quad \checkmark$$

c) i) $\angle PNQ = \angle PMQ = 90^\circ$ [$\angle$'s in same segment] $\checkmark$

ii) Join QA and AR.

Let $\angle RPA = \alpha = \angle RQA$

[$\angle$'s in same segment in cyclic quad PQAR] $\checkmark$

Also $\angle NPM = \alpha = \angle NQM$

[$\angle$'s in same segment in cyclic quad PQMN] $\checkmark$

Also since $\angle HMQ = \angle AMQ = 90^\circ$

and since QM is a common side

then $\triangle QHM \equiv \triangle QAM$ [AAS] $\checkmark$

Hence $HM = MA$ [corres sides in cong Δ 's]

$$d) i) \frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{1}{\frac{dx}{d\theta}}$$

$$\therefore m = \frac{dy}{dx} = b \cos \theta \times \frac{1}{-a \sin \theta} = -\frac{b \cos \theta}{a \sin \theta} \quad \checkmark$$

$$\therefore y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$ay \sin \theta - ab \sin^2 \theta = -b x \cos \theta + ab \cos^2 \theta$$

$$\therefore b x \cos \theta + ay \sin \theta = ab (\sin^2 \theta + \cos^2 \theta)$$

$$b x \cos \theta + ay \sin \theta = ab \quad [\div ab] \text{ to get } \quad \checkmark$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1.$$

ii) M has coordinates $(0, b \sin \theta)$

In the equation in (i) let $x = 0$ to give

the coordinates of B $\left(0, \frac{b}{\sin \theta}\right)$. $\checkmark$

$$\therefore OM \cdot OB = |b \sin \theta| \times \frac{b}{|\sin \theta|} = b^2 \quad \checkmark$$

$$6a) \quad m_{PQ} = \frac{\frac{c}{q} - \frac{c}{p}}{cq - cp} = \frac{c(p-q)}{pq} \times \frac{1}{c(q-p)} = \frac{-1}{pq}$$

$$\text{Since } \angle PRQ = 90^\circ \text{ then } m_{PR} \times m_{QR} = -1 \quad \checkmark$$

$$\therefore \frac{-1}{pr} \times \frac{-1}{qr} = -1 \rightarrow r^2 = -\frac{1}{pq} \quad \checkmark$$

$$\text{Now } xy = c^2 \rightarrow y' = -\frac{c^2}{x^2}$$

$$\text{Hence at } R \text{ the gradient of the tangent} = \frac{-1}{r^2}$$

$$\therefore \text{ at } R \text{ the gradient of the normal} = r^2 \quad \checkmark$$

$$\text{but } r^2 = -\frac{1}{pq} = m_{PQ} \quad \checkmark$$

$$\therefore PQ \text{ parallel to the normal at } R.$$

$$bi) \quad \text{Let } r \text{ be the radius of a typical slice}$$

$$\therefore r + 1 = x \rightarrow r = x - 1 \quad \checkmark$$

$$\text{Now } \Delta V = \pi r^2 h = \pi(x-1)^2 \Delta y \quad \checkmark$$

$$ii) \quad \text{When } x = 1, y = 2 - 1 = 1 \quad \checkmark$$

$$\therefore V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^1 \pi(x-1)^2 \Delta y$$

$$\text{Now } -y = x^2 - 2x$$

$$\therefore 1 - y = x^2 - 2x + 1 \quad \checkmark$$

$$\therefore 1 - y = (x-1)^2$$

$$\text{hence } \therefore V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^1 \pi(1-y) \Delta y$$

$$iii) \quad V = \pi \int_0^1 1 - y \, dy$$

$$= \pi \left[y - \frac{y^2}{2} \right]_0^1 \quad \checkmark$$

$$= \frac{\pi}{2} u^3 \quad \checkmark$$

$$c) i) \quad (a-b)^2 > 0 \quad (a \neq b)$$

$$\therefore a^2 - 2ab + b^2 > 0 \quad \checkmark$$

$$\therefore a^2 + b^2 > 2ab \text{ -----} \boxed{1}$$

$$ii) \quad \text{Similarly } b^2 + c^2 > 2bc \text{ -----} \boxed{2}$$

$$\text{and } c^2 + a^2 > 2ca \text{ -----} \boxed{3} \quad \checkmark$$

Now $\boxed{1} + \boxed{2} + \boxed{3}$ gives

$$2(a^2 + b^2 + c^2) > 2(ab + bc + ca) \quad \checkmark$$

$$\therefore a^2 + b^2 + c^2 > ab + bc + ca$$

iii) Using the result in (ii)

$$\text{Let } a \rightarrow ab; \quad b \rightarrow bc; \quad c \rightarrow ca.$$

$$\therefore a^2 b^2 + b^2 c^2 + c^2 a^2 > ab \cdot bc + bc \cdot ca + ca \cdot ab \quad \checkmark$$

$$a^2 b^2 + b^2 c^2 + c^2 a^2 > abc(a + b + c) \quad \checkmark$$

$$\therefore \frac{a^2 b^2 + b^2 c^2 + c^2 a^2}{a+b+c} > abc$$

$$7a) \quad P(x) = 3x^3 - 11x^2 + 8x + 4$$

$$P'(x) = 9x^2 - 22x + 8$$

$$P'(x) = 0 \text{ for stationary points.}$$

$$\therefore (9x-4)(x-2) = 0 \rightarrow x = 2 \text{ or } x = \frac{4}{9} \quad \checkmark$$

$$\text{Since } P(2) = 24 - 44 + 16 + 4 = 0$$

$$\text{then } x = 2 \text{ is the double root.} \quad \checkmark$$

$$\therefore P(x) = (x-2)^2 (3x+1) \quad \checkmark$$

$$b) \quad \text{Let } P(x) = \tan^{-1} x - x + \frac{1}{3} x^3$$

$$\therefore P'(x) = \frac{1}{1+x^2} - 1 + x^2 \quad \checkmark$$

$$= \frac{1 - (1+x^2) + x^2(1+x^2)}{1+x^2}$$

$$= \frac{x^4}{1+x^2} > 0 \text{ for } x > 0 \quad \checkmark$$

$$\therefore P(x) \text{ is an increasing function for } x > 0$$

$$\text{and since } P(0) = 0 \text{ then } \tan^{-1} x > x - \frac{1}{3} x^3 \quad \checkmark$$

$$c) i) \quad \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2x^3 - 10x$$

$$\therefore \frac{1}{2} v^2 = \frac{x^4}{2} - 5x^2 + c \quad \checkmark$$

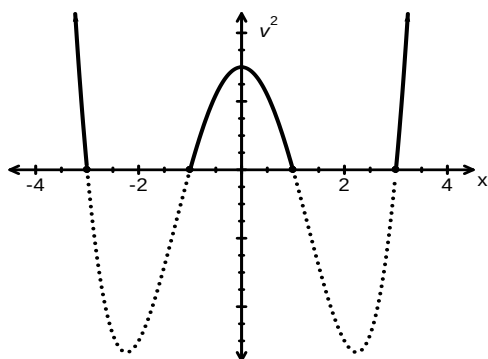
$$\text{when } x = 0, v = u \rightarrow c = \frac{1}{2} u^2$$

$$\therefore \frac{1}{2} v^2 = \frac{x^4}{2} - 5x^2 + \frac{1}{2} u^2 \quad \checkmark$$

$$\therefore v^2 - u^2 = x^4 - 10x^2$$

7cii) If $u = 3$ then $v^2 - 9 = x^4 - 10x^2$

$$\begin{aligned}\therefore v^2 &= x^4 - 10x^2 + 9 \\ &= (x^2 - 1)(x^2 - 9) \\ &= (x - 1)(x + 1)(x - 3)(x + 3) \quad \checkmark\end{aligned}$$



Since $v^2 \geq 0$ for motion to exist then
 $(x - 1)(x + 1)(x - 3)(x + 3) \geq 0$
 and since the particle starts at $x = 0$ with $v = 3$ it
 is moving to the right. $\checkmark$

At $x = 1$, $v = 0$ and $\ddot{x} = -8$ and so the particle
 will move to the left until it reaches $x = -1$
 where $v = 0$ and $\ddot{x} = 8$. This means the
 particle will then move to the right until
 $v = 0$ again at $x = 1$. $\checkmark\checkmark$

Thus it oscillates in the interval $-1 \leq x \leq 1$.

iii) Not SHM since $\ddot{x} \neq -n^2(x - b)$ $\checkmark$

iv) When $u = 6$, $v^2 = x^4 - 10x^2 + 36$
 $= (x^2 - 5)^2 + 11$

$\therefore v^2 > 0$ for all x in the domain, hence the
 particle will never stop. $\checkmark$

It moves to the right with decreasing velocity
 until it passes $x = \sqrt{5}$ (with $v = \sqrt{11}$) and
 continues to the right with increasing velocity. $\checkmark$

8 a) i) Let ω be a solution to $z^3 = 1$

$$\begin{aligned}\therefore z^3 - 1 &= 0 \rightarrow (z - 1)(z^2 + z + 1) = 0 \\ \text{but as } \omega &\text{ is a non real root, then} \\ \omega^2 + \omega + 1 &= 0. \quad \checkmark\end{aligned}$$

ii) Now $\omega^3 = 1$, hence

$$\begin{aligned}(2 - \omega)(2 - \omega^2)(2 - \omega^4)(2 - \omega^5) &\text{ reduces to} \\ (2 - \omega)(2 - \omega^2)(2 - \omega)(2 - \omega^2) &\quad \checkmark \\ = (2 - \omega)^2(2 - \omega^2)^2 & \\ = [(2 - \omega)(2 - \omega^2)]^2 & \\ = [4 - 2\omega^2 - 2\omega + \omega^3]^2 & \\ = [5 - 2(\omega^2 + \omega)]^2 & \\ = [5 - 2 \times -1]^2 = 49 &\quad \checkmark\end{aligned}$$

$$\begin{aligned}8 \text{ b) i) } \cos(3\theta) &= \cos(\theta + 2\theta) \\ &= \cos \theta \cos 2\theta - \sin \theta \sin 2\theta \\ &= \cos \theta (2 \cos^2 \theta - 1) - \sin \theta 2 \sin \theta \cos \theta \quad \checkmark \\ &= 2 \cos^3 \theta - \cos \theta - 2 \cos \theta (1 - \cos^2 \theta) \\ &= 2 \cos^3 \theta - \cos \theta - 2 \cos \theta + 2 \cos^3 \theta \quad \checkmark \\ &= 4 \cos^3 \theta - 3 \cos \theta.\end{aligned}$$

$$\begin{aligned}\text{ii) } x^3 - 3x &= \sqrt{2} \text{ and let } x = 2 \cos \theta \text{ to obtain} \\ 8 \cos^3 \theta - 6 \cos \theta &= \sqrt{2} \\ \therefore 4 \cos^3 \theta - 3 \cos \theta &= \frac{1}{\sqrt{2}} \\ \therefore \cos 3\theta &= \frac{1}{\sqrt{2}} \quad \checkmark\end{aligned}$$

$$\therefore 3\theta = \frac{\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \dots$$

$$\therefore \theta = \frac{\pi}{12}, \frac{7\pi}{12}, \frac{9\pi}{12}, \dots \quad \checkmark$$

$$\text{hence } x = 2 \cos \frac{\pi}{12}, 2 \cos \frac{7\pi}{12}, 2 \cos \frac{9\pi}{12} \quad \checkmark$$

iii) The sum of the roots of $x^3 - 3x - \sqrt{2} = 0$
 is given by $-\frac{b}{a} = -\frac{0}{1} = 0$.

$$\begin{aligned}\therefore 2 \cos \frac{\pi}{12} + 2 \cos \frac{7\pi}{12} + 2 \cos \frac{9\pi}{12} &= 0 \quad \checkmark \\ \therefore \cos \frac{\pi}{12} + \cos \frac{7\pi}{12} + \cos \frac{9\pi}{12} &= 0\end{aligned}$$

c) i) $y = x^3 - 3px + q$, $y' = 3x^2 - 3p$, $y'' = 6x$

For stat pts $y' = 0 \rightarrow 3(x^2 - p) = 0$

$$\therefore x = \pm\sqrt{p} \quad \checkmark$$

$$\text{When } x = \sqrt{p}, y = p\sqrt{p} - 3p\sqrt{p} + q = q - 2p\sqrt{p}$$

$$\text{When } x = \sqrt{p}, y'' = 6\sqrt{p} > 0 \rightarrow \text{minimum.} \quad \checkmark$$

$$\begin{aligned}\text{When } x = -\sqrt{p}, y &= -p\sqrt{p} + 3p\sqrt{p} + q \\ &= q + 2p\sqrt{p}\end{aligned}$$

$$\text{When } x = -\sqrt{p}, y'' = -6\sqrt{p} < 0 \rightarrow \text{maximum.} \quad \checkmark$$

ii) Since $x^3 - 3px + q = 0$ has 3 distinct non
 zero real roots then the turning points must
 be on either side of the x - axis. $\checkmark$

$$\therefore \text{minimum} \times \text{maximum} < 0 \quad \checkmark$$

$$\begin{aligned}\therefore (q + 2p\sqrt{p})(q - 2p\sqrt{p}) &< 0 \\ q^2 - 4p^2 \cdot p &< 0 \\ \text{hence } q^2 &< 4p^3 \quad \checkmark\end{aligned}$$